KTH ROYAL INSTITUTE
OF TECHNOLOGY

Artificial Intelligence and Multi Agent Systems DD2438

Lecture 4

Assignment 3

If your partner is slacking, let me know!

Module 3 (4 weeks, 100h of work)

w13 Wed 2026-03-25 08:00 - 10:00 W42

w14 Tue 2026-03-31 10:00 - 12:00 W42

w15 (EASTER)

w16 Tue 2026-04-14 10:00 - 12:00 W42 (Competition1)

w17 Tue 2026-04-21 10:00 - 12:00 W42 (Competition2)

Eat 1 pill (Executable)

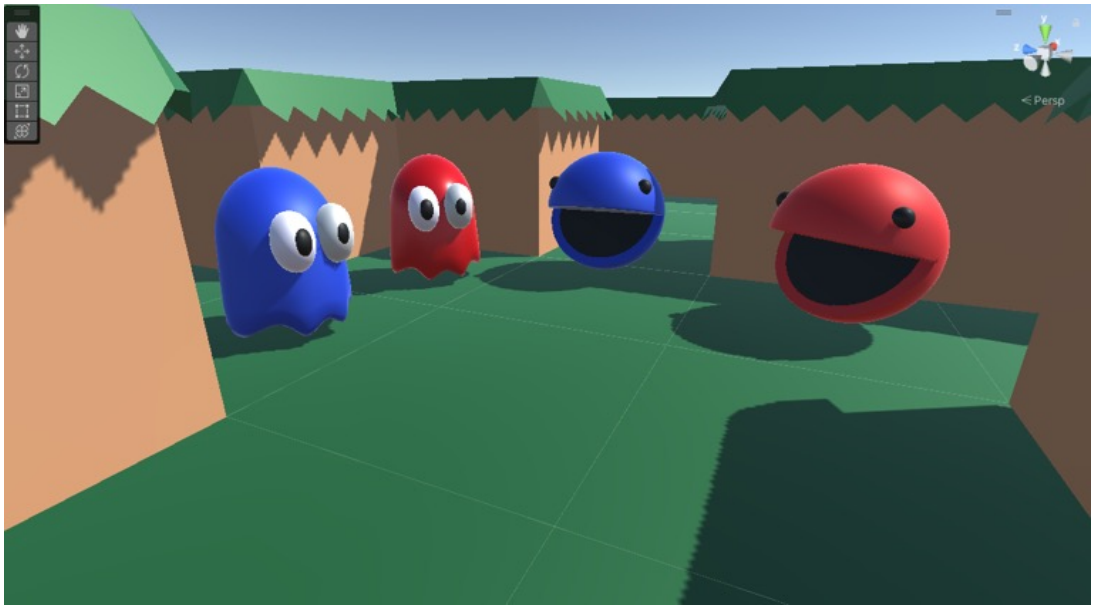
First competition

Second competition

The Problem of Assignment 3

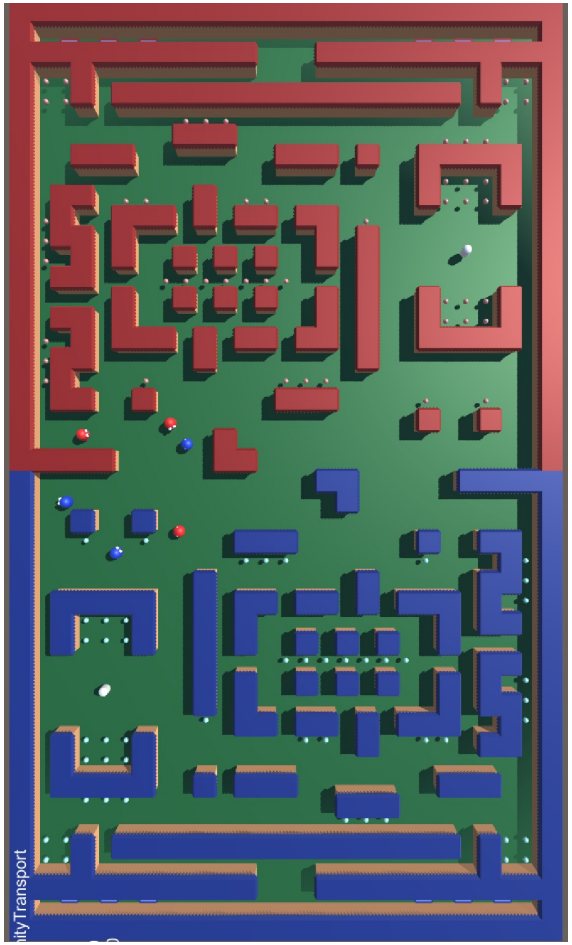
- Drone-Dynamics Pacman Capture the Flag

- (demonstration soon…)

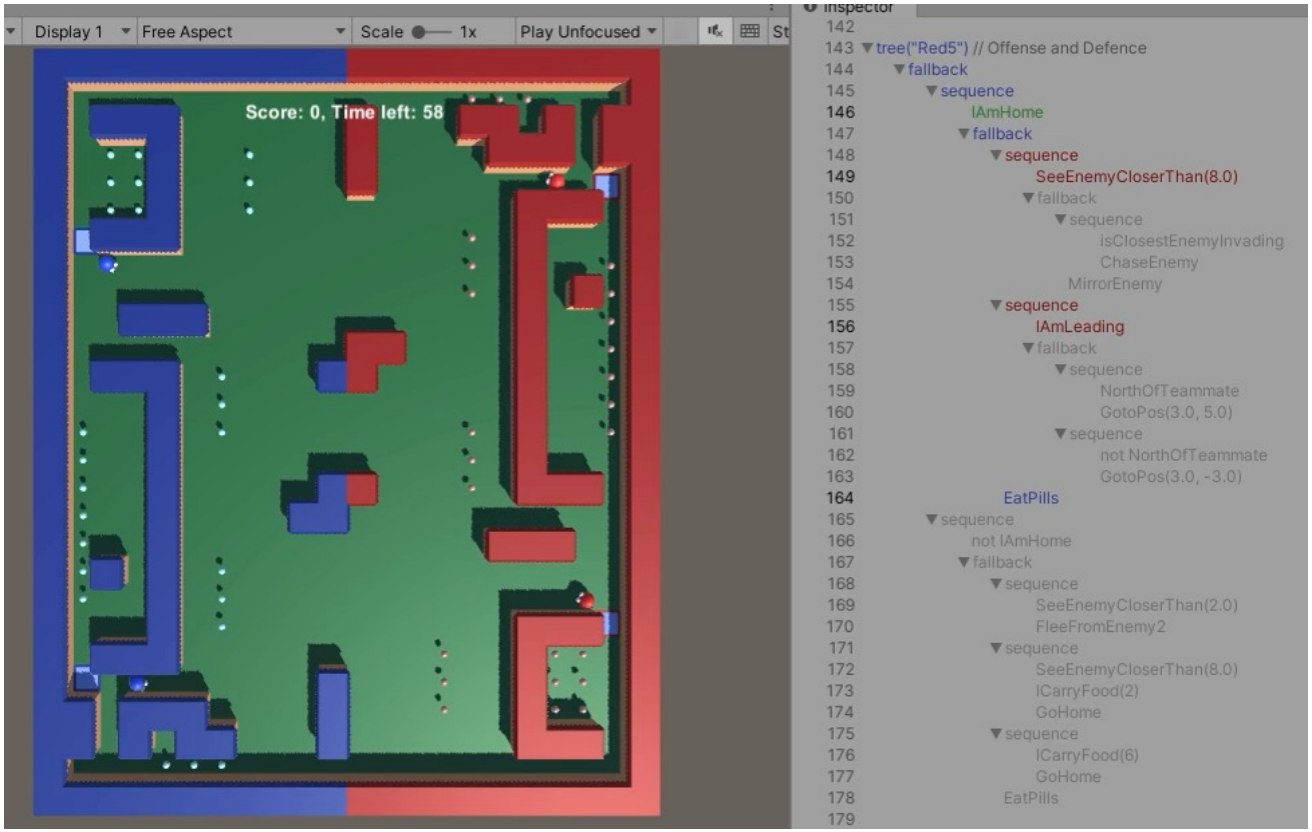


Pac Man Capture the Flag

- Symmetric maze (blue/red sides)
- Ghost on home side, Pac Man on other
- Eat the pills of the opponents
- Bring back home
- Winner has more pills
- Super-pill
  - Pac man can eat ghosts for 10s
  - Ghost die from super-pill
- All killed agents will
  - Drop what ever pills they have eaten
  - Re-spawn on their own side

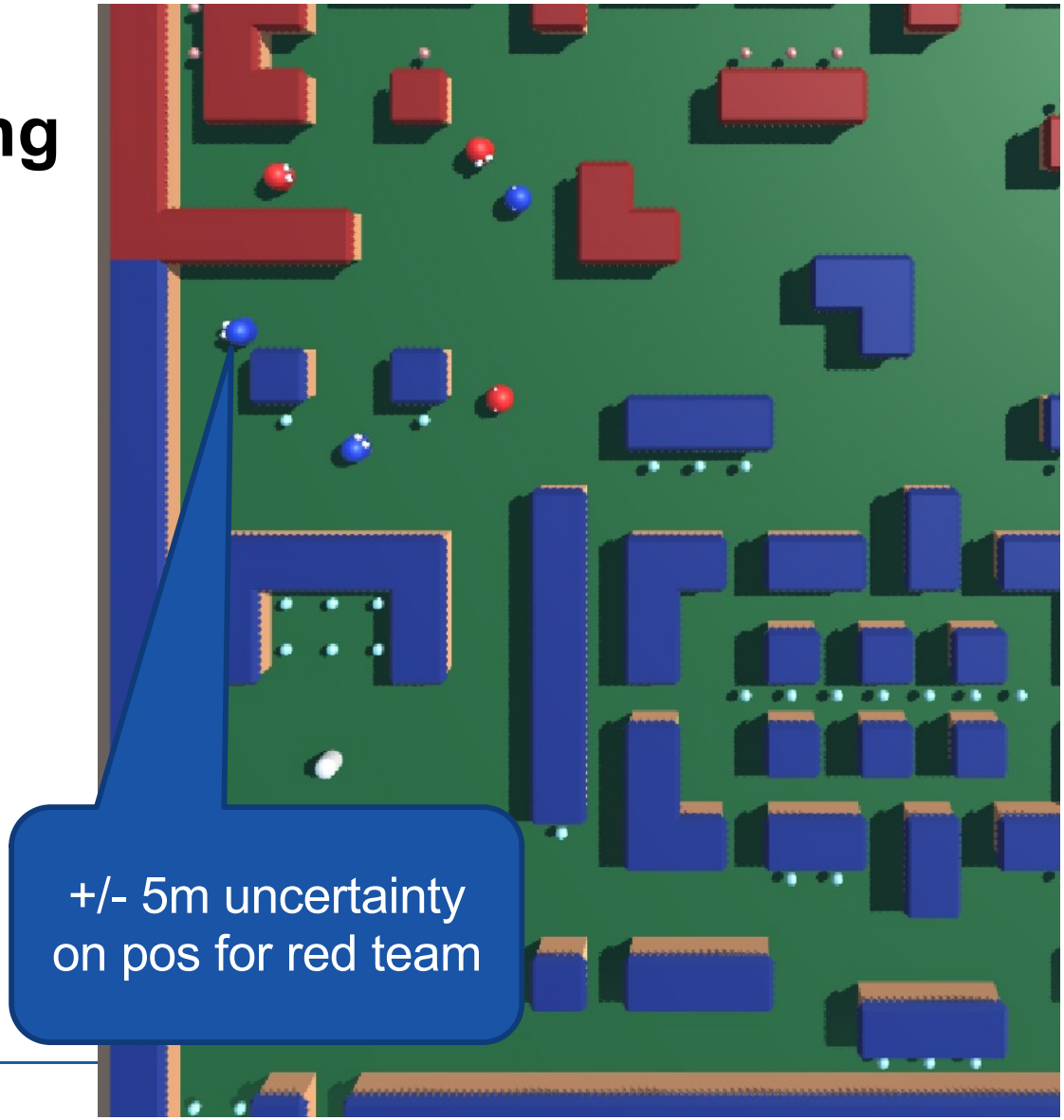


Example game



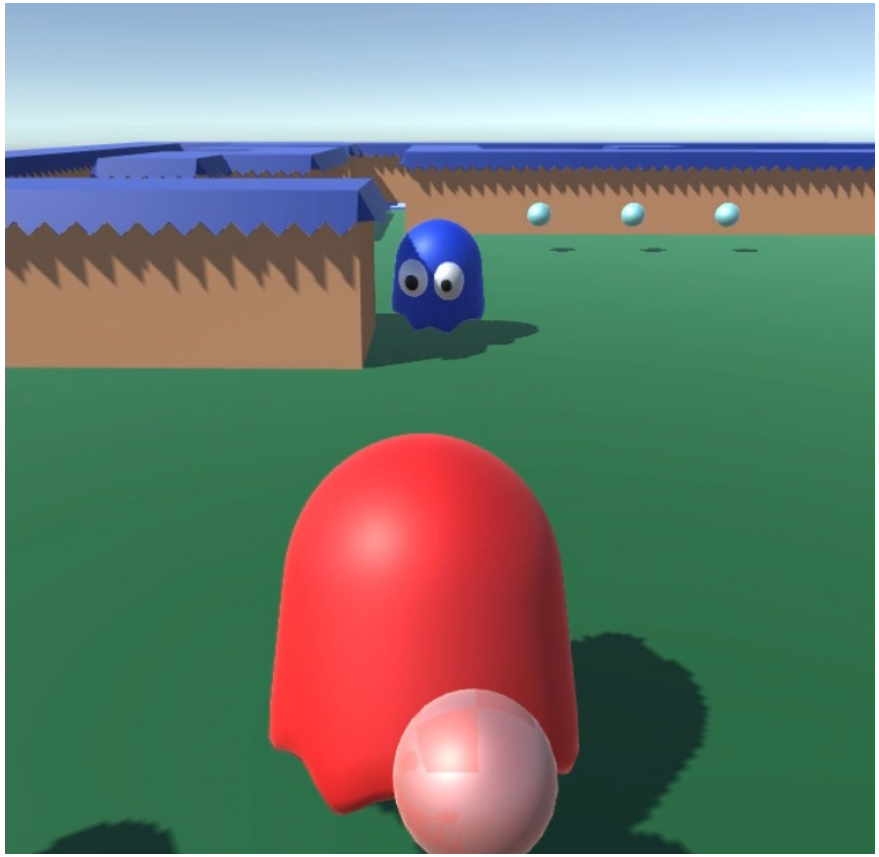
Pacman CTF Sensing

- You get the position of enemies within line of sight
- You can “hear” enemies moving above 0.3m/s
  - With a +/- 5m error (resampled every 1s)
- You know the location of all pills
  - if a pill gets eaten, there is an enemy there
- You know how many pills each enemy is carrying



Decisions to be made

- Defense or Attack?



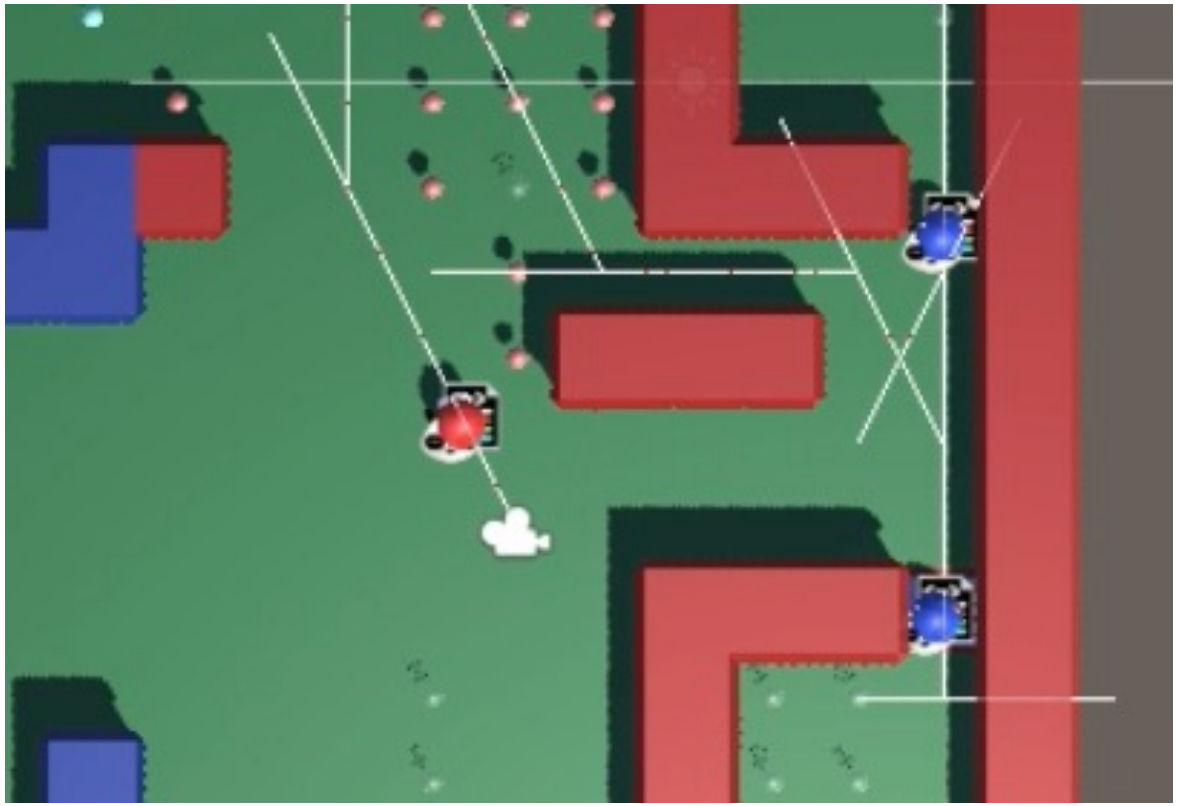
Decisions to be made

- Blue
  - Safety or Pills?
- Red
  - Aggressive of covering of exits?

Decisions to be made

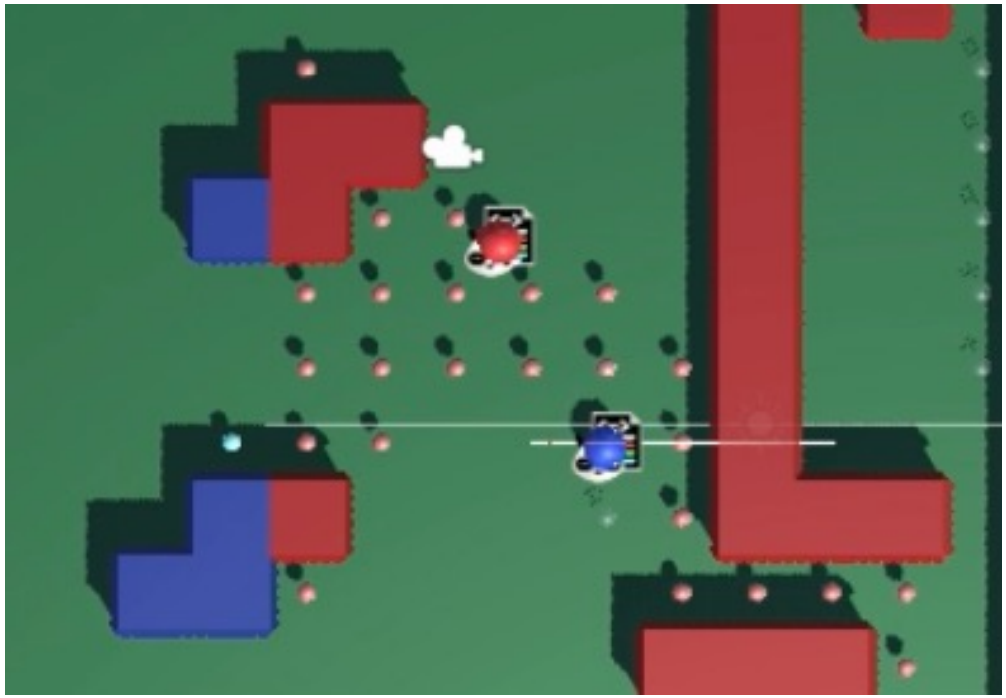
- Red cannot chase both blue
  - Who carries most pills?

- Blue can use this



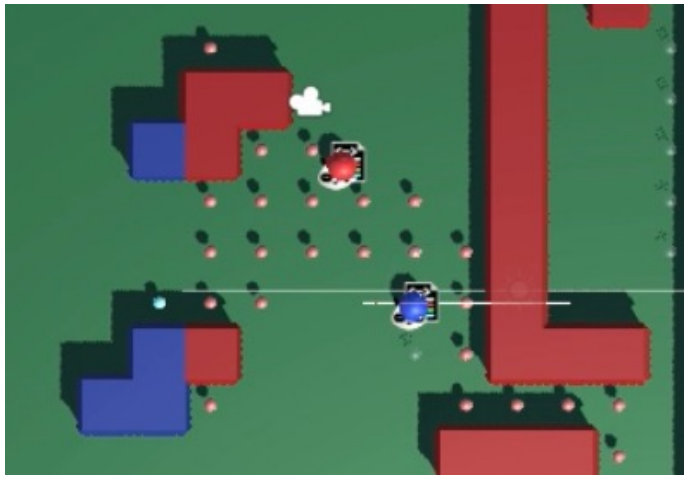
Decisions to be made

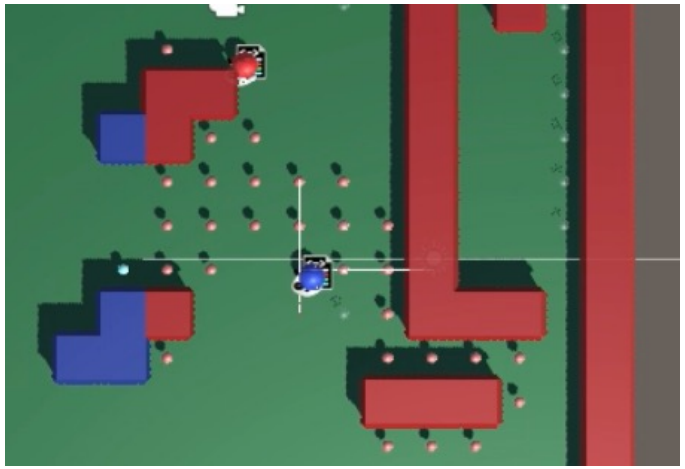
- Shortest way home is not always safest



Decisions to be made

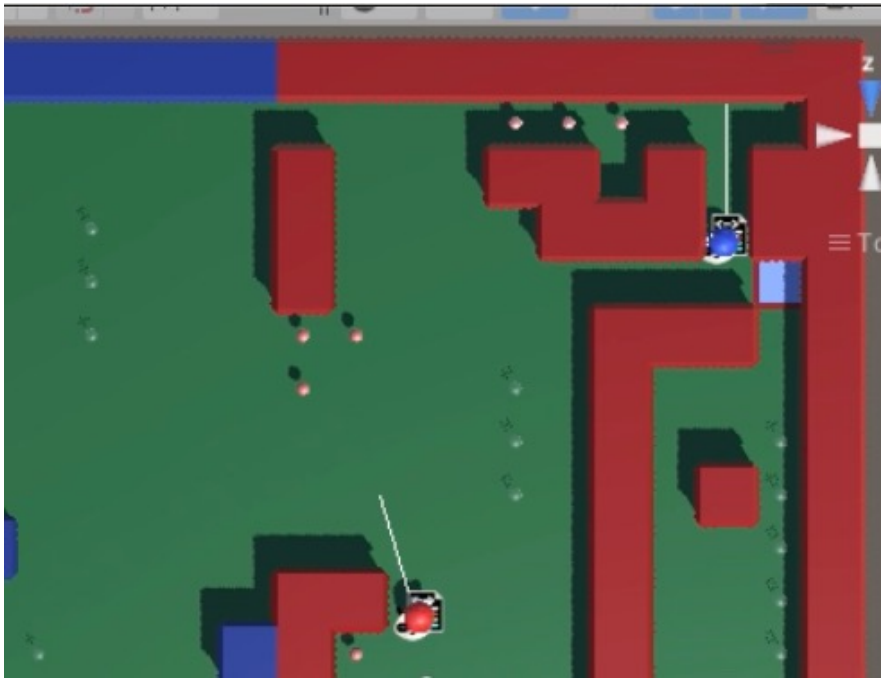
- Shortest way home is not always safest (part 2)



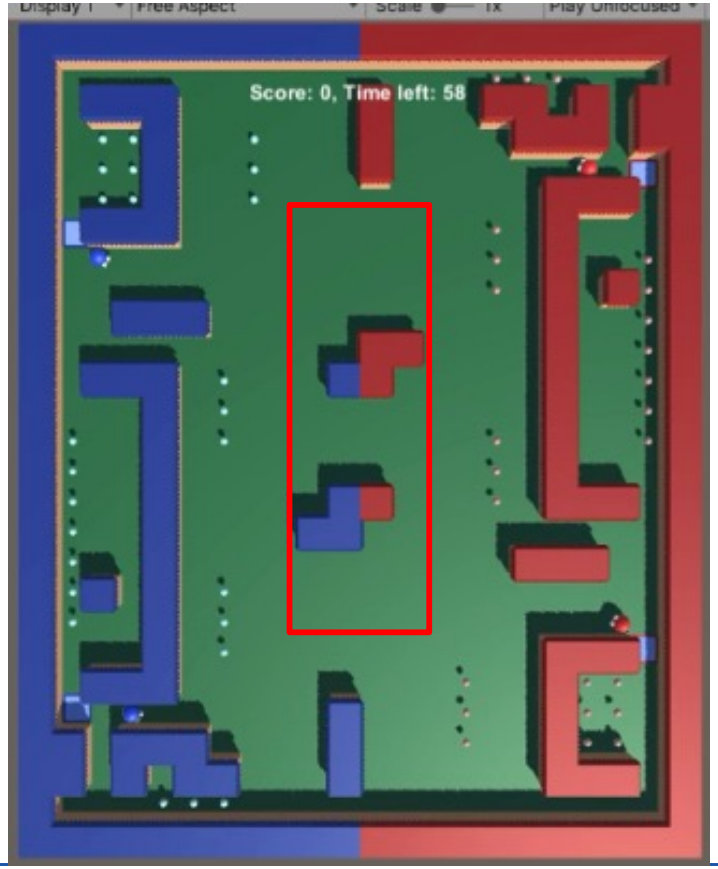


Decisions to be made

- Go fast or slow?



Border Battle will be Important

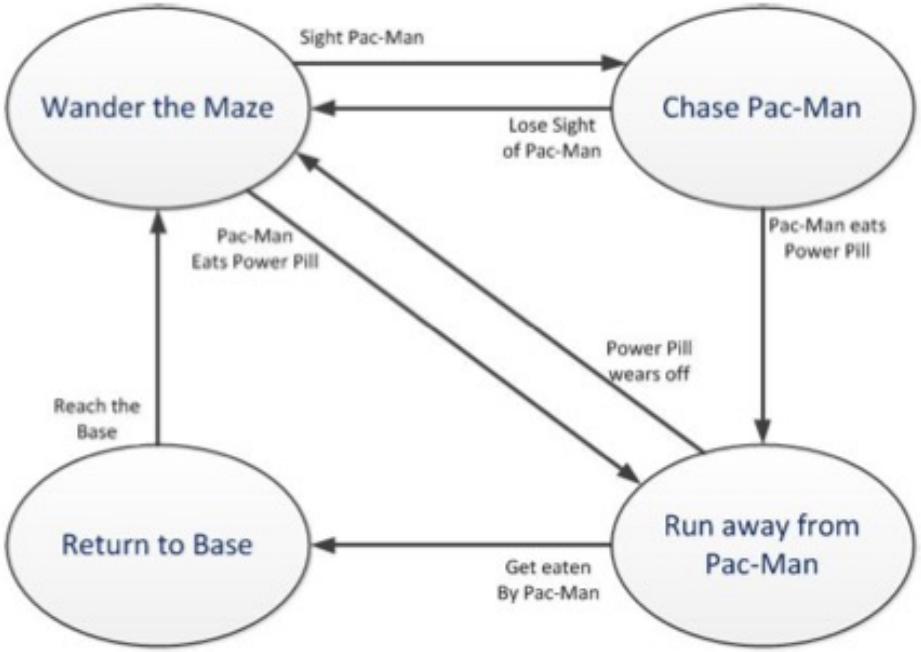


- This is where stolen pills end up
- … RL opportunities…
- … or other designs…

5 ideas

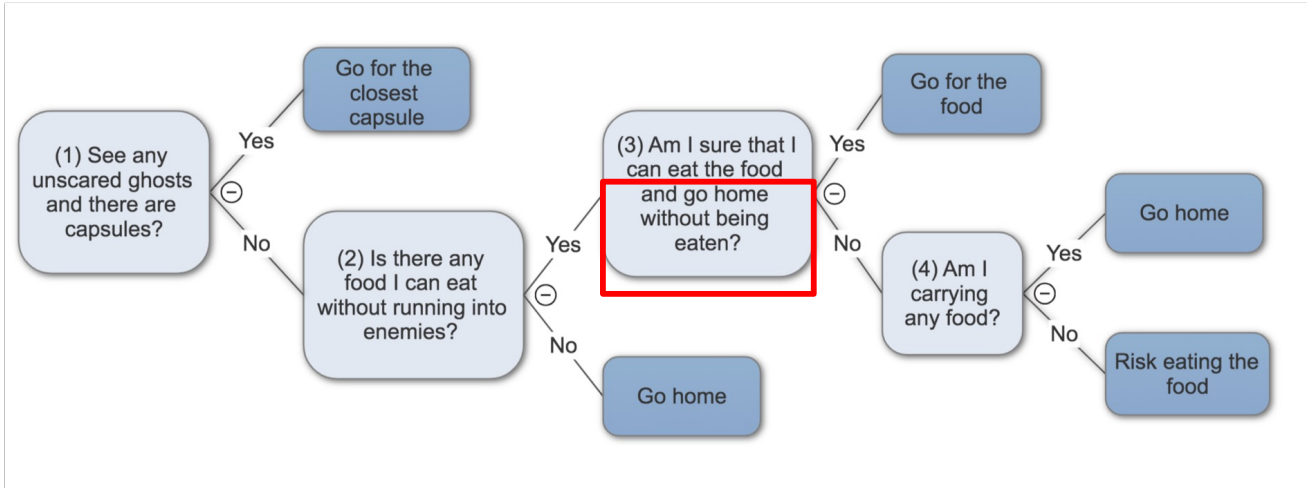
- Deciding what to do…
- Voronoi Partitioning
- Running in Circles
- Tree search
- Particle Filters

Deciding what to do: State Machines



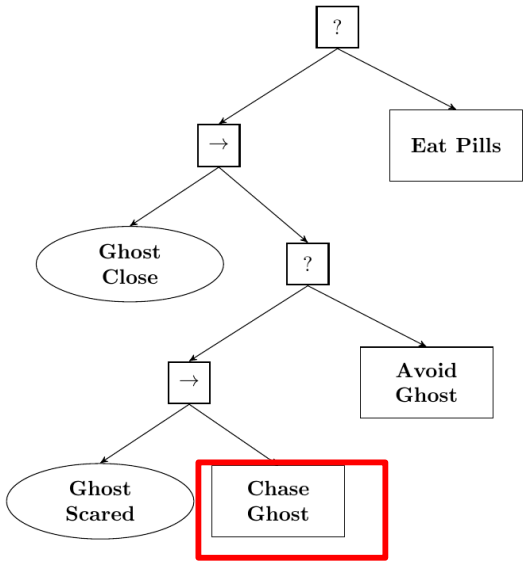
Here ”what to do” depends on “what am I doing” & “world state (good when you have to “finish something first”)

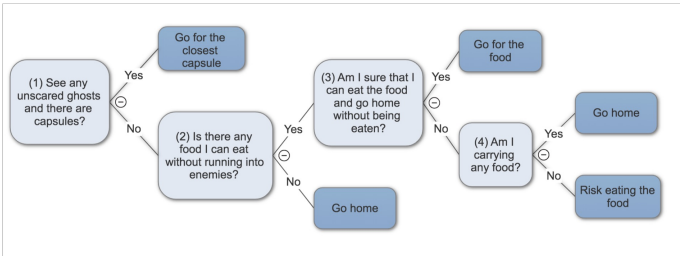
Deciding what to do: Decision Tree



Here ”what to do” depends on “world state”

Deciding what to do: Behavior Tree





Here ”what to do” depends on “world state”
Actions can “Fail”
(More complex than Decision Trees)

5 ideas

- Deciding what to do…
- Voronoi Partitioning
- Running in Circles
- Tree search
- Particle Filters

Pac Man and Voronoi Partitions

Normal Voronoi diagram →

- All “closest points”
- All point that can be reached “first”
- If
  - Power pill
  - Safe side
- …is in your Voronoi cell
- … then you are good

- What about obstacles?

The Ambulance Positioning Problem

- The closest ambulance goes to the victim
- We want to minimize the longest response time for 112 calls
- Street map is a graph
- Wang et al., *A Distributed Convergent Solution to the Ambulance Positioning Problem on a Streetmap Graph*. *IFAC* 2014.

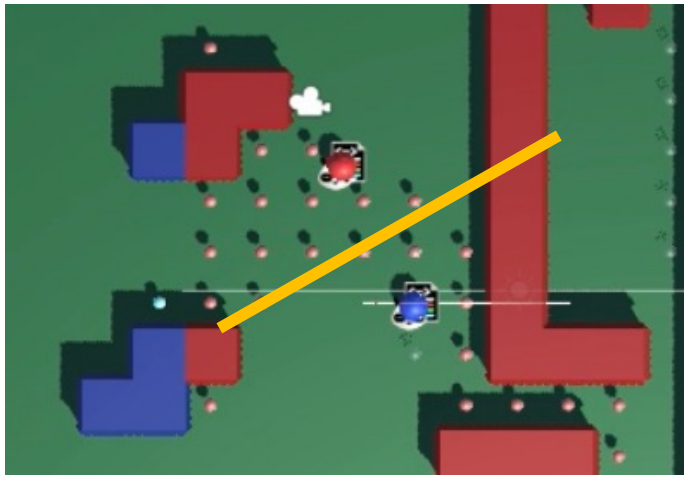
Initial

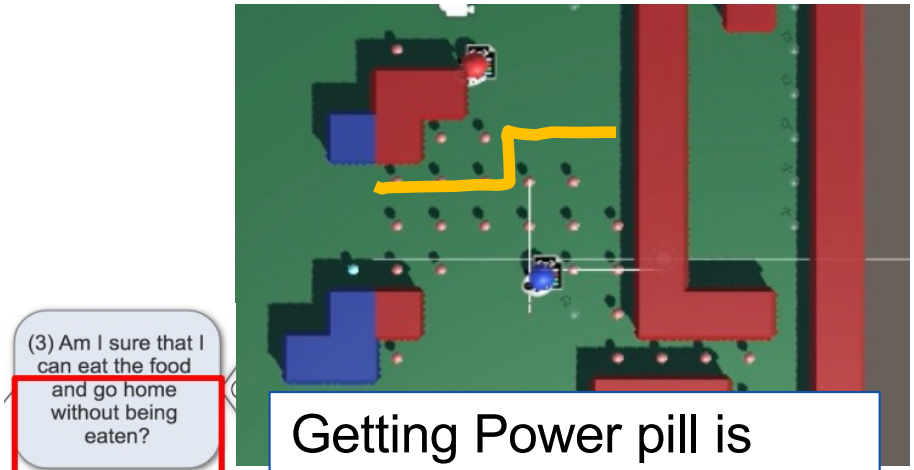
with 4 agents

with 3 agents

Decisions to be made

- Shortest way home is not always safest (part 3)
- Voronoi-partitions
  - continuous
    - > *more accurate in open spaces*
  - discrete
    - > *handles obstacles*
    - > *2 A* searches in same graph (check smallest)*
    - > *Floyd Warshall algorithms (all pairs shortest path)*





(3) Am I sure that I can eat the food and go home without being eaten?

Getting Power pill is enough to get home!

5 ideas

- Deciding what to do…
- Voronoi Partitioning
  - Finding safe paths home
- Running in Circles
  - If there are no safe paths home
- Tree search
- Particle Filters

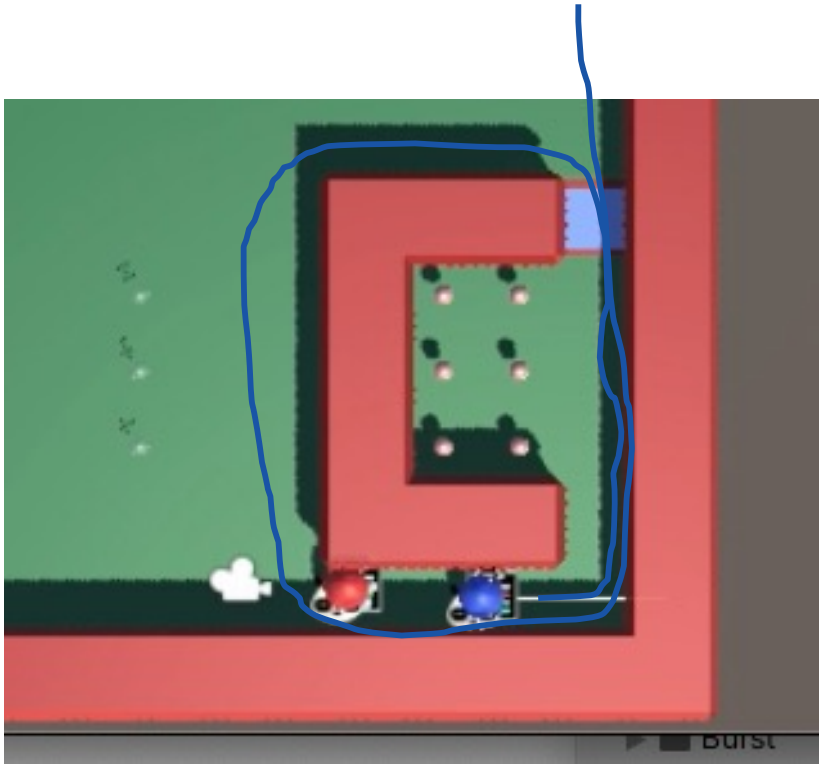
Running in circles

When chased:

- Loops are good
- Dead ends are bad

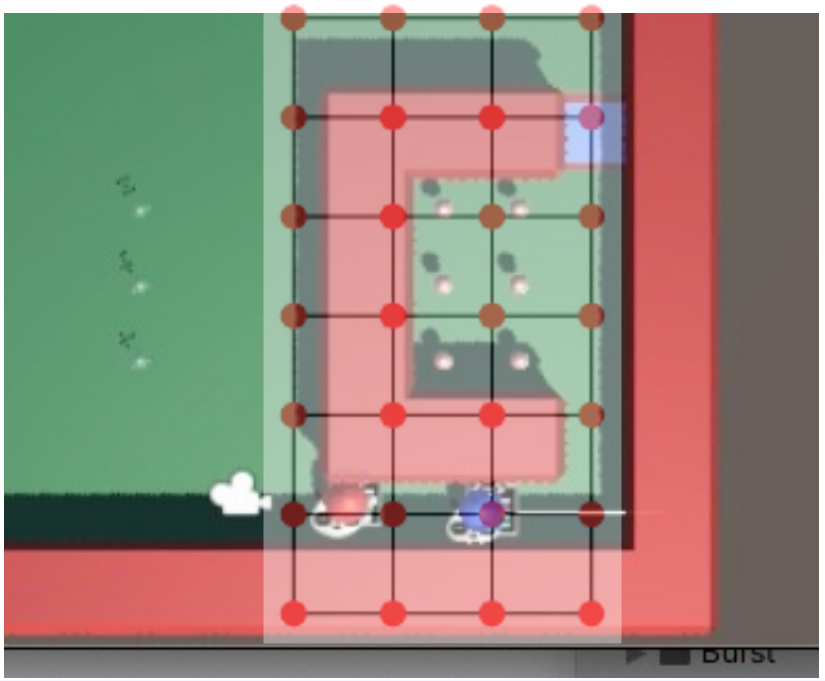
How to find shortest loop?

- Use “line of sight perimeters” from Nilsson, et al., *Optimal Positioning of Surveillance UGVs*. IEEE IROS *2008*.



Finding Shortest Loop (around obstacle)

- Given graph
- Draw line
  - from inside obstacle
  - Through player pos
  - To edge of world
- Remove all edges cut by line
- Pick removed edge closest to player (A,B)
- Find shortest path between A and B in resulting graph



Running this loop, 2 enemies are needed to get you!

Nilsson, et al., *Optimal Positioning of Surveillance UGVs*. IEEE IROS *2008*

5 ideas

- Deciding what to do…
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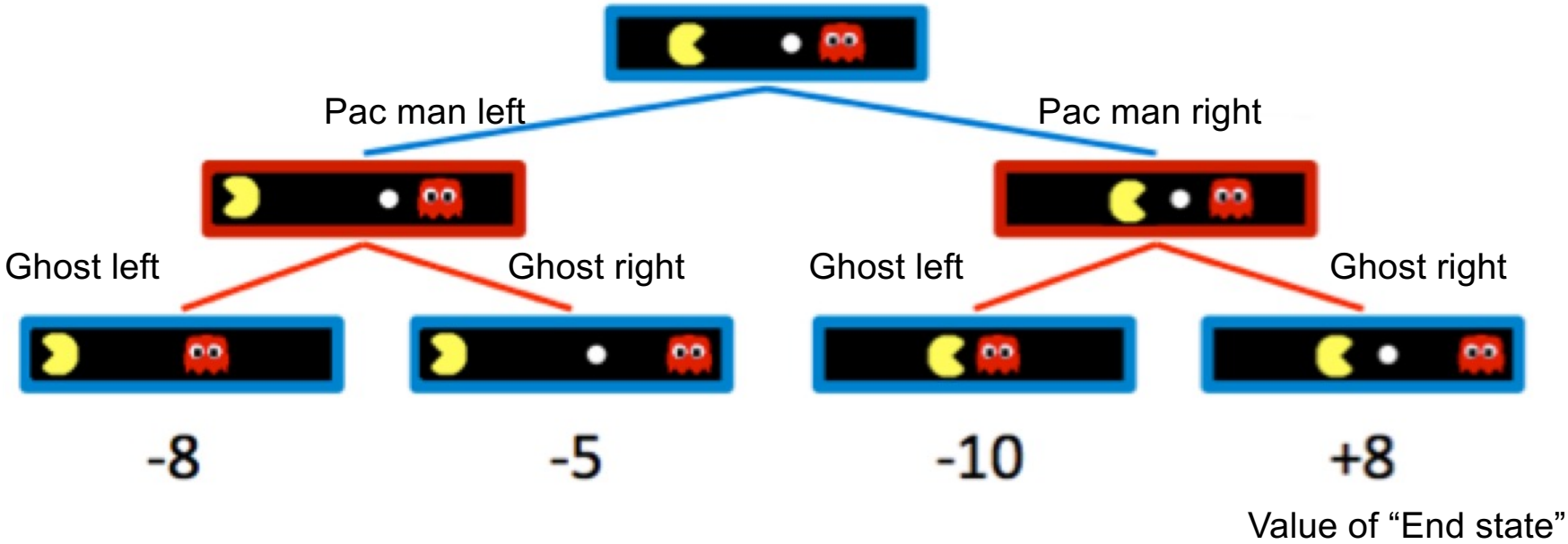
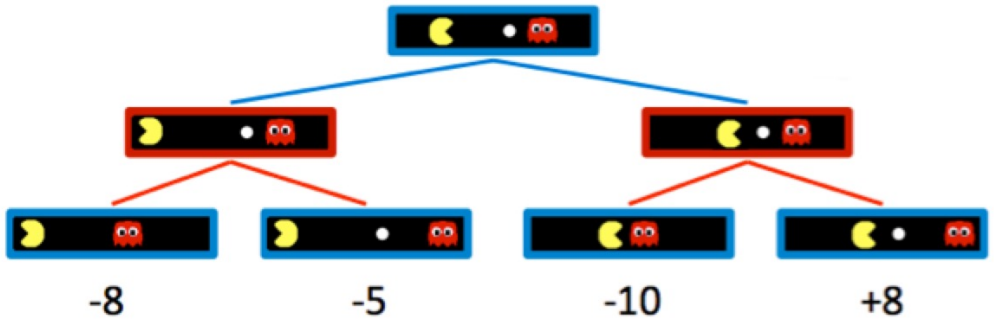
Tree search

- Minimax Algorithm
- AlfaBeta Pruning
- Monte Carlo Tree Search



Can use Voronoi Graph for “Value” (Can I get back home safely?)

Monte Carlo Tree search



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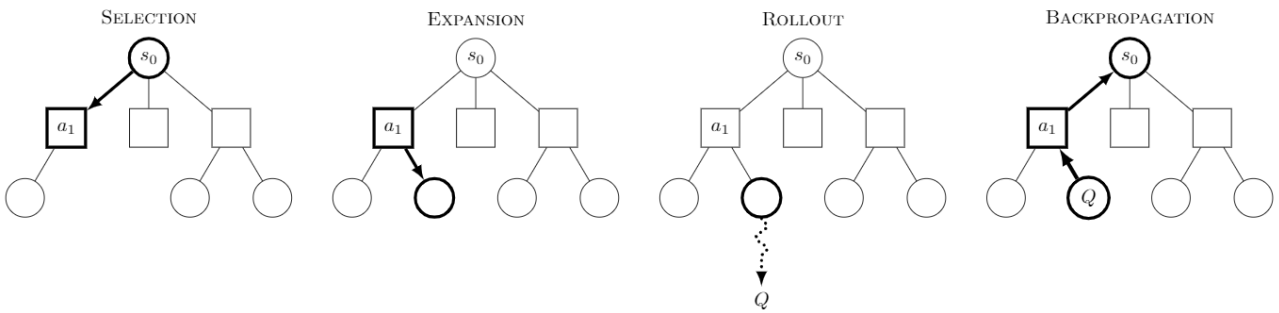

5 ideas

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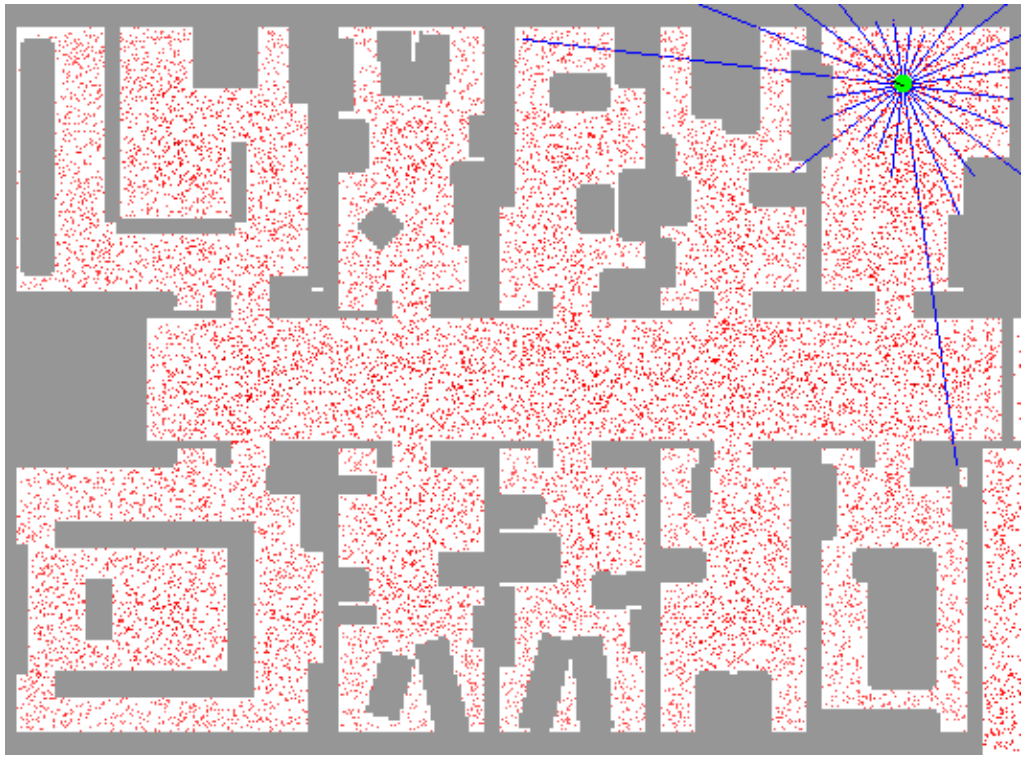
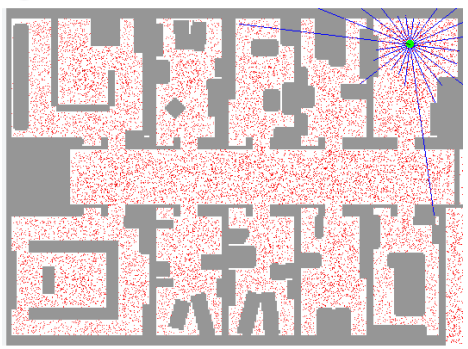
Particle Filters

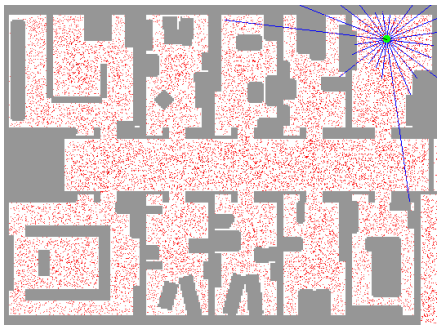


Given a map, and sensor readings: Where am I?



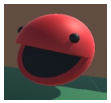
Particle Filters

1. Start with a bunch of particles that represent where you think you are right now.
2. Prediction step: propagate each particle through the motion model.
3. Correction step: if a particle is unlikely to produce the measurement we obtain, “just get rid of it!”
4. Repeat from 2



where are the opponents?

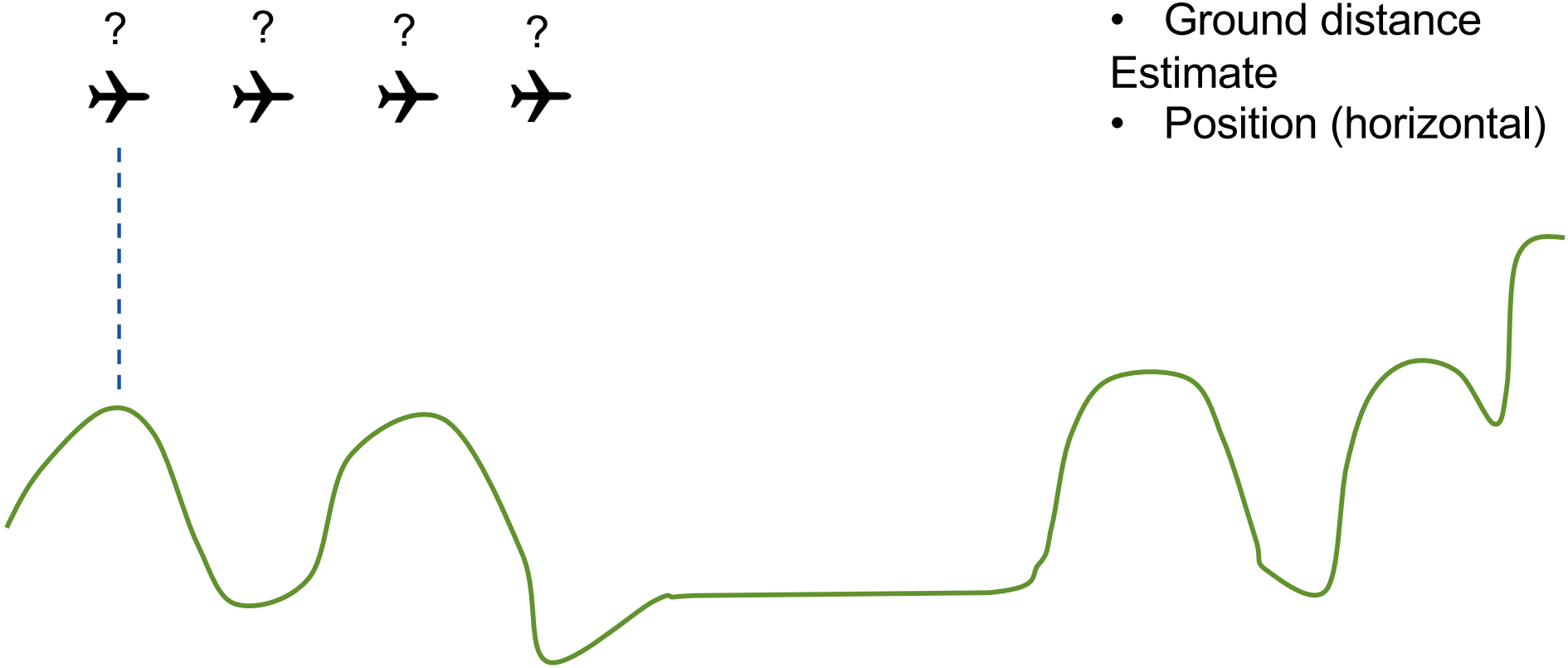
- see/hear



Particle Filters

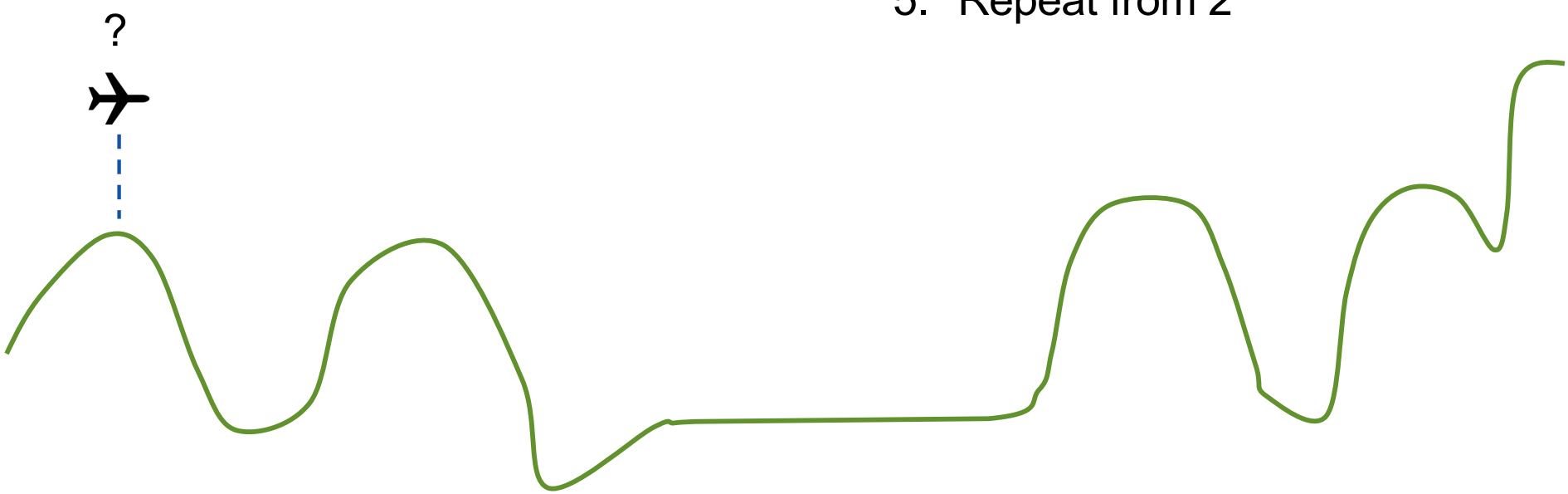
Given
- Map
- Altitude above sea
- Ground distance

Estimate
- Position (horizontal)



Particle Filters

1. Start (Uniform distribution)
2. Prediction step: propagate each particle through the motion model.
3. Check Likelihood → Weights
4. Correction step: Resample (to remove unlikely samples)
5. Repeat from 2



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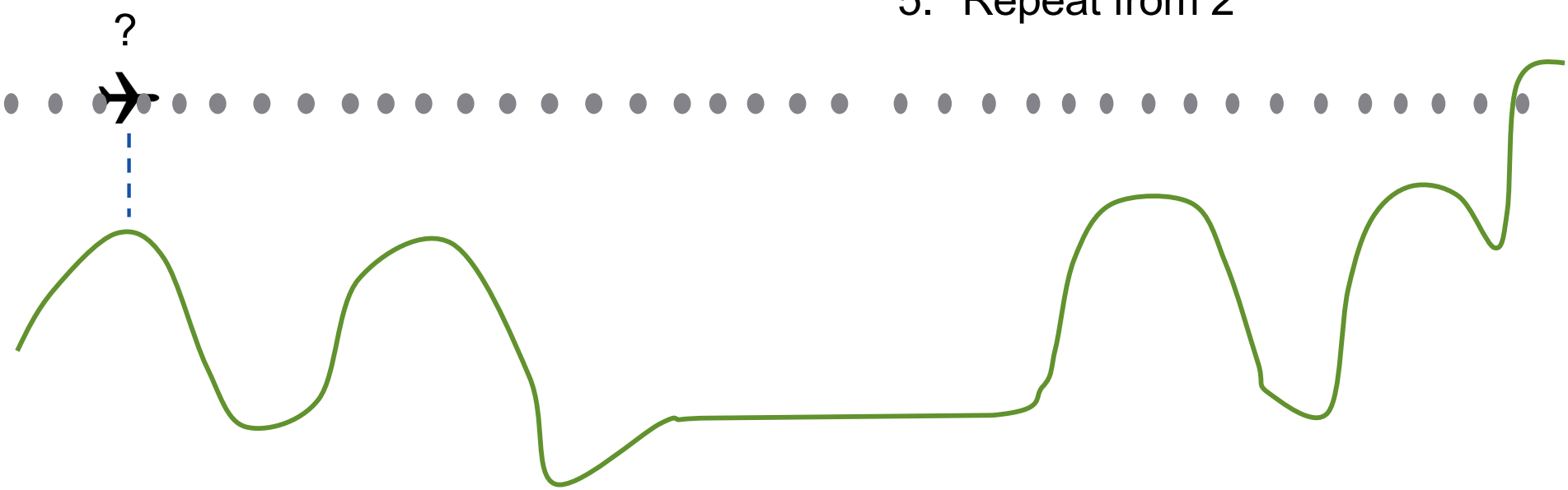

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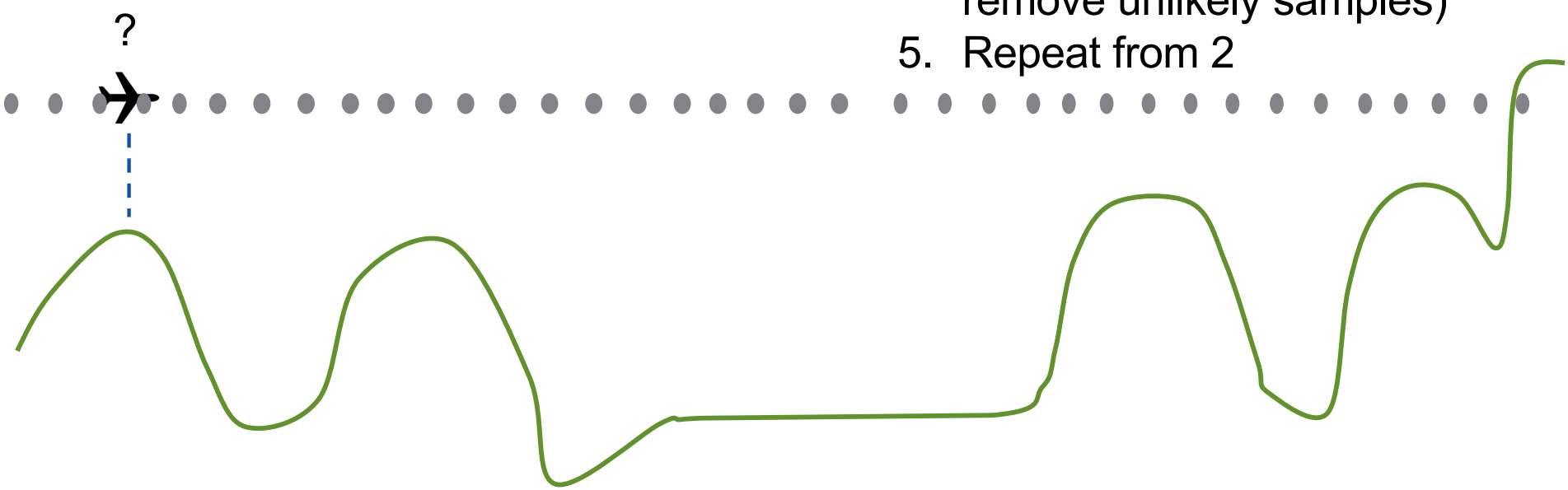

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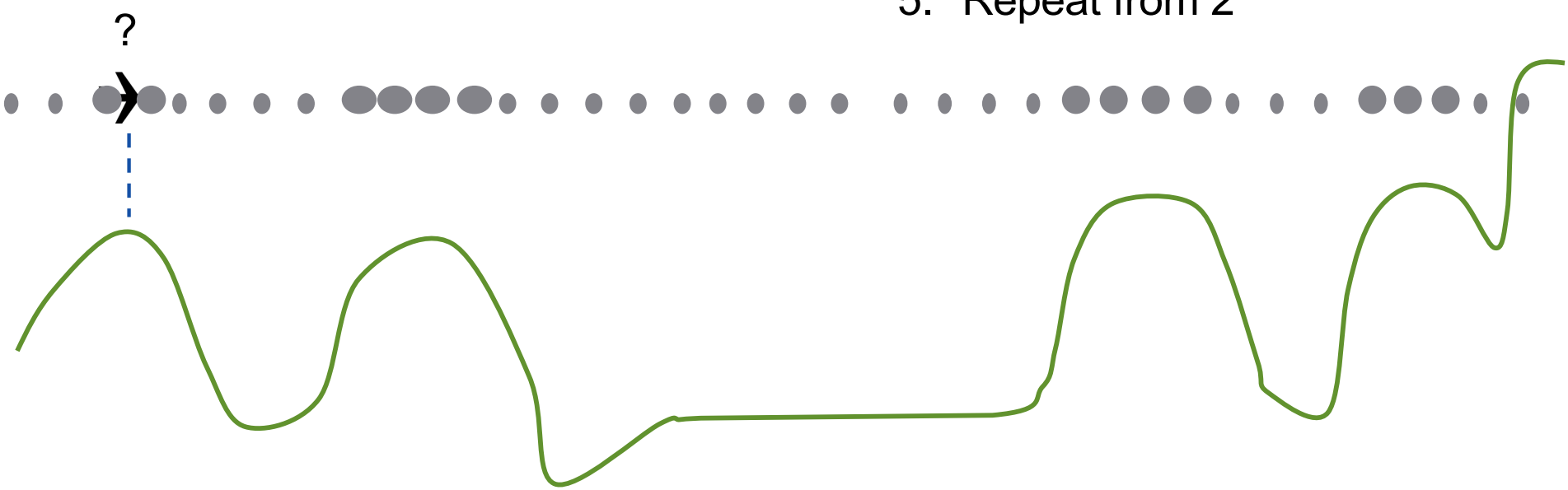

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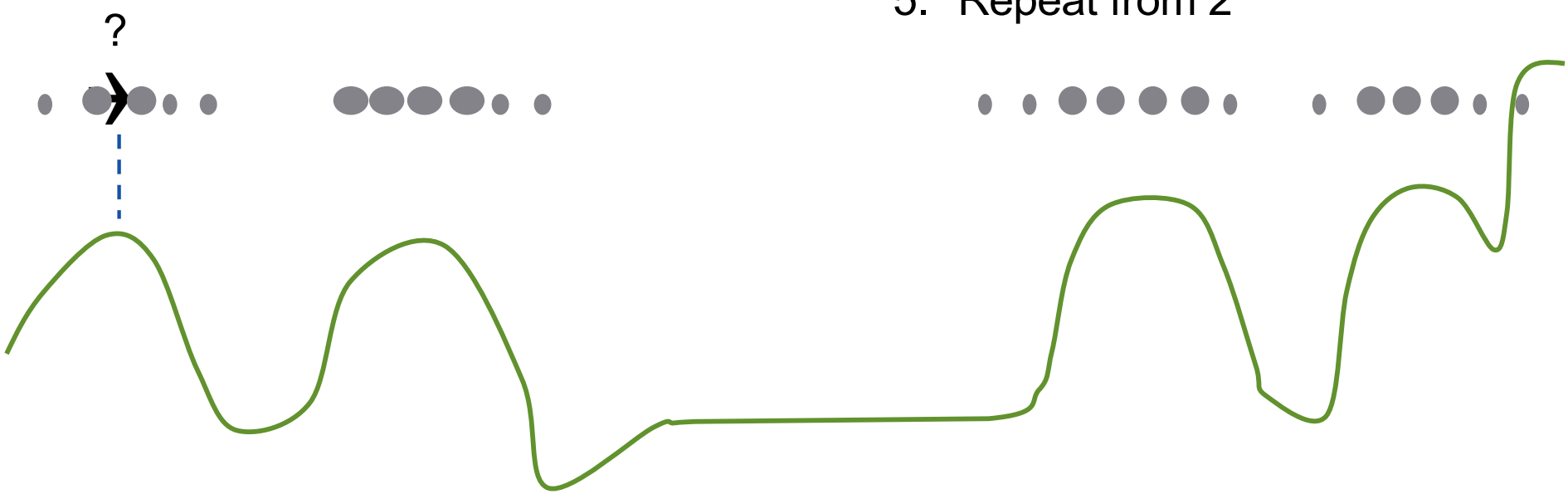

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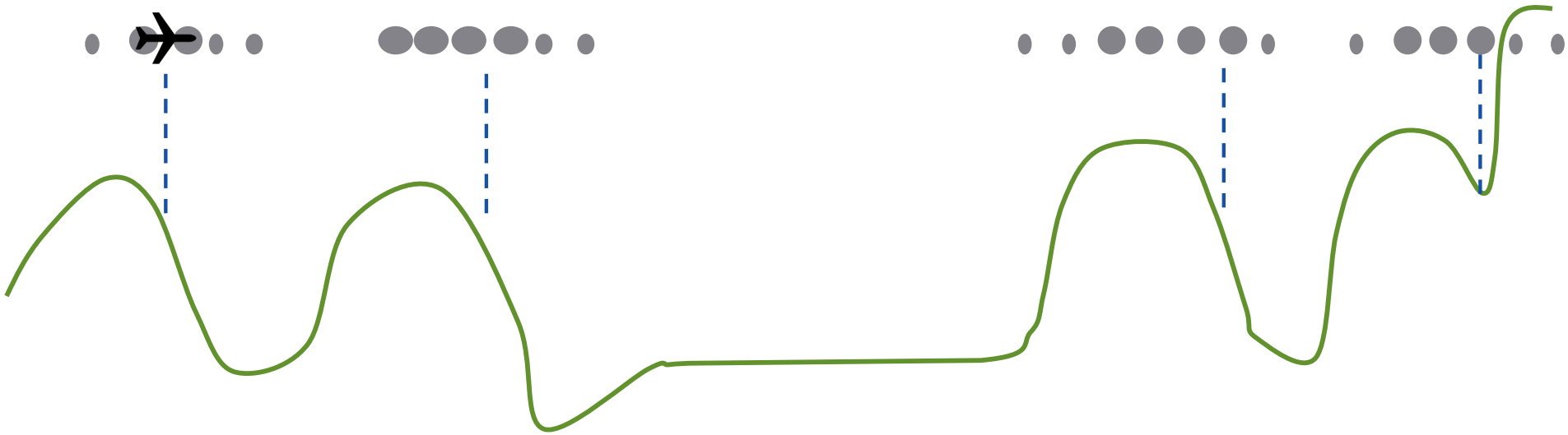

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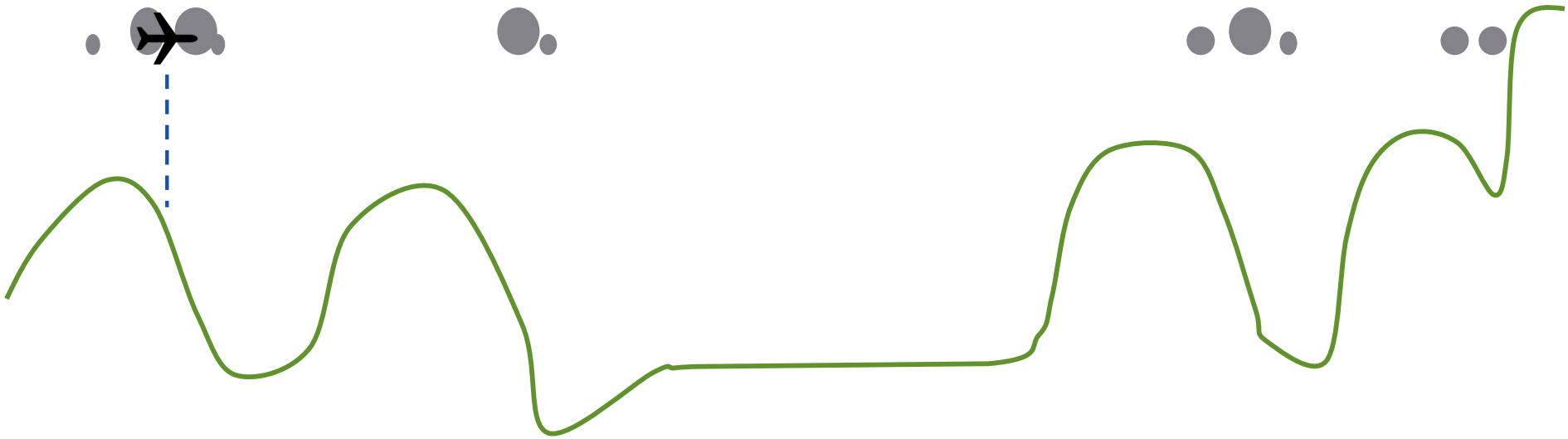

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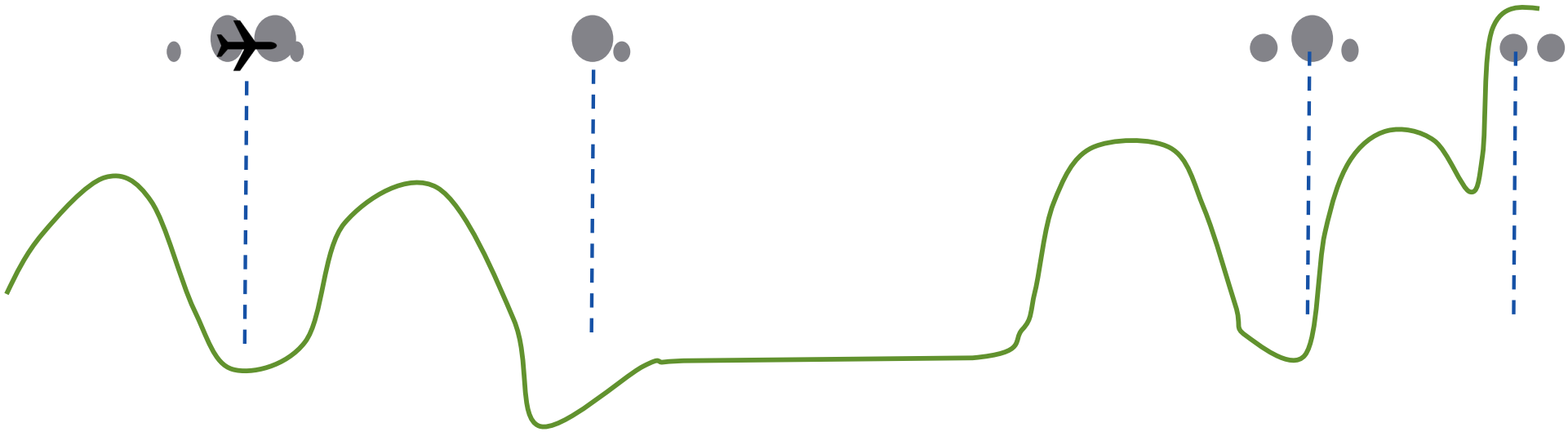

Following Andreas Svensson, Uppsala University

Particle Filters

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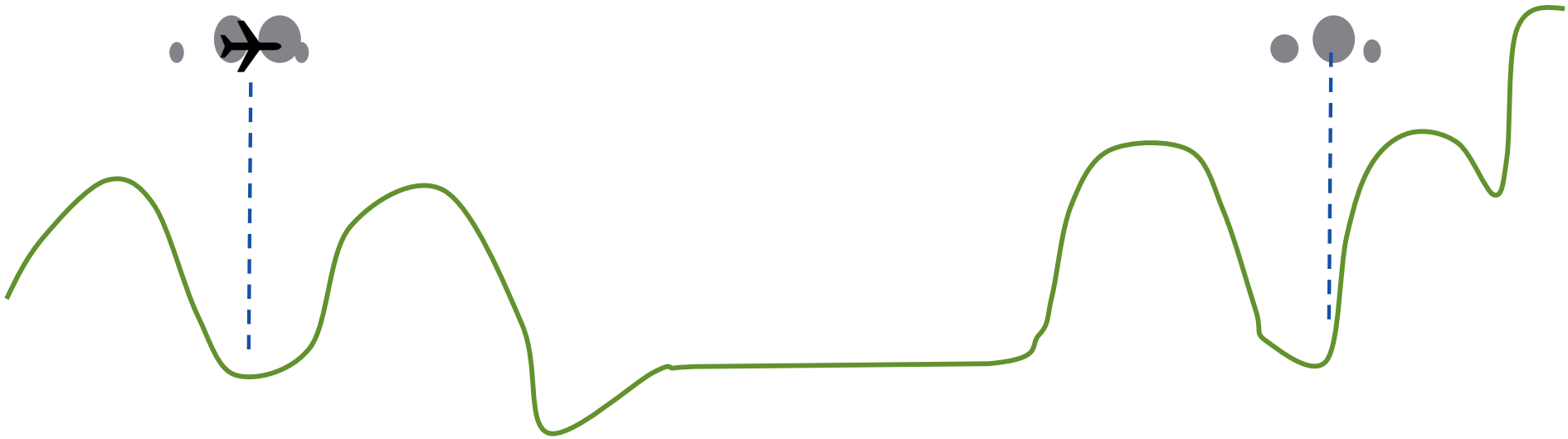

Following Andreas Svensson, Uppsala University

5 ideas

- Deciding what to do…
- Voronoi Partitioning
  - Finding safe paths home
- Running in Circles
  - If there are no safe paths home
- Tree search
- Particle Filters

References

- Monte Carlo tree search
- Minimax Games
- Particle Filters
- Wang et al., *A Distributed Convergent Solution to the Ambulance Positioning Problem on a Streetmap Graph*. *IFAC* 2014
- Nilsson, et al., *Optimal Positioning of Surveillance UGVs*. IEEE IROS *2008*.
- Rohlfshagen, Philipp, et al. "Pac-man conquers academia: Two decades of research using a classic arcade game." IEEE Transactions on Games 10.3 (2017).
- Pepels, Tom, Mark HM Winands, and Marc Lanctot. "Real-time monte carlo tree search in ms pac-man." IEEE Transactions on Computational Intelligence and AI in games 6.3 (2014): 245-257.

- ... and whatever you can find in Google Scholar... (and elsewhere)